

Learning Objectives

- Explain what open data is and identify major reliable sources of it.
- Describe how data is obtained through an API, and explain the importance of terms of use and source citation.

Some of the most powerful data in the world is free to use — population figures from the World Bank, global statistics from the UN, competition datasets on Kaggle. But "free" is not the same as "no rules." To rely on open data you need to know where to get it, how to fetch it (a download in a browser, or an API that returns data directly to your program), and how you are allowed to use it. Checking the terms of use and citing your source are what turn borrowed data into trustworthy analysis.

? Guiding question: Where does reliable open data come from, how do programs fetch it, and what rules govern its use?

📌 Point!

1. Using Open Data

(1) To use open data in analysis, it is necessary to understand where to obtain it, how to obtain it, and how to use it. By correctly observing these points, reliable analysis can be performed.

(2) Open data comes from a variety of sources, such as data from international organizations, data from government agencies, and datasets published by researchers and companies. It is important to choose data sources with high reliability that match the purpose of the analysis.

2. Major Open Data Sources

(1) The following are typical, internationally widely used providers of open data.

Provider	Description	Main uses
(World Bank Open Data)	Statistics on economy, population, education, and health for countries around the world	International comparisons and analysis of social/economic trends in each country
(UN Data)	International statistics published by the United Nations and its related agencies	Analysis of global issues such as population, development, and environment
(Kaggle)	A platform that runs data analysis competitions; datasets in various fields are made public	Acquisition of data for machine learning study and research

(2) On these platforms, data can be searched for and downloaded from a browser, and in many cases data can also be obtained through APIs.

3. Obtaining Data via APIs

(1) **(API (Application Programming Interface))**…rules for how pieces of software interact with each other. In web services, APIs are also widely used as a

Key Fact:
Open data comes from official providers (World Bank Open Data, UN Data) and platforms such as Kaggle, whose reliability varies; get it from a browser or through an API (often returning JSON); every dataset has terms of use to follow and a source to cite.

Key Concepts:

- open data
- World Bank Open Data / UN Data / Kaggle
- API / endpoint / JSON
- terms of use
- source citation

KEY TERMS

open data — data made public for anyone to use.
World Bank Open Data — country statistics on economy, population, and more.



Open data is public and free to reuse.

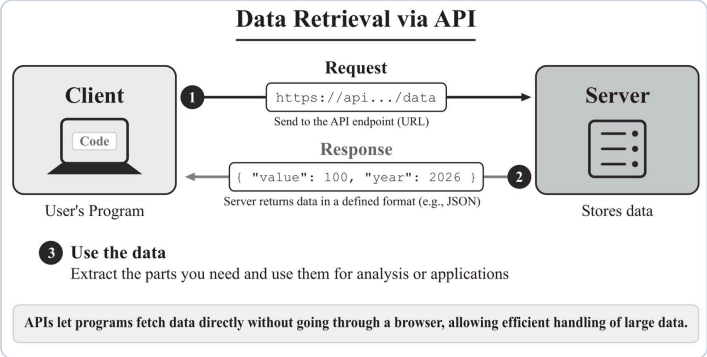
means of obtaining data. By using APIs, necessary data can be obtained directly without going through a browser, allowing large amounts of data to be handled efficiently.

(2) Obtaining data via API proceeds according to the following flow.

① The client (the user's program) sends a request to the API endpoint (the URL that is the destination of the request).

② The server processes the request and returns the requested data in a predetermined format.

③ The client receives the response, extracts the necessary parts, and uses them.



Data retrieval via API: request, response, and use

(3) The data format most widely used for data returned via APIs today is **(JSON (JavaScript Object Notation))**. JSON is a notation that represents data as combinations of names and values; it is easy for humans to read and easy to handle in various programming languages.

(4) When using an API, it is important to **(follow the terms of use)**, not send excessive requests in a short time, and confirm the scope of allowed use of the obtained data.

4. Data Use and Source Citation

(1) Even with open data, the scope of free use is set for each dataset. Always confirm the **(terms of use)** of the data before using it.

(2) When using data, it is required to **(cite the source)**. The source generally includes the data provider, the name of the data, the date of acquisition, and the URL.

(3) Citing the source is important both as an indication of the reliability of the data and to allow others to verify the analysis.

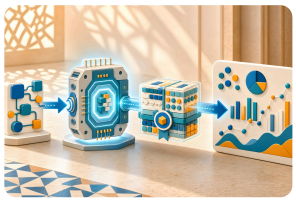
✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A World Bank Open Data publishes statistics on economy, population, education, and so on for countries around the world.

KEY TERMS

API — rules for how software interacts, used to fetch data.
endpoint — the URL a request is sent to.
JSON — a common name/value data format.



An API delivers fresh data straight into analysis.

PAUSE & THINK

"Open" data still has rules — what two things must you check before you use it?



Use data responsibly and credit its source.

B Using an API, necessary data can be obtained directly without going through a browser.

C As long as data is open data, it can be used freely without checking the terms of use.

D When using data, it is required to cite the provider, the name of the data, the date of acquisition, the URL, and so on as the source.

(2) Match each description (a–c) with the most appropriate open data provider from the choices below (A–C).

a Publishes statistics on economy, population, education, and so on for countries around the world.

b Provides international statistics published by the United Nations and its related agencies.

c A platform that runs data analysis competitions.

[Choices] **A** UN Data **B** World Bank Open Data **C** Kaggle

PAUSE & THINK

In an API call, who sends the request and who returns the data?

✓ Solution

(1)

A Correct description of World Bank Open Data. Therefore, ○.

B Correct description of an API. Therefore, ○.

C Even for open data, the terms of use differ for each dataset, so they must always be checked before use. Therefore, ×.

D Correct content of source citation. Therefore, ○.

(2)

a: Statistics on countries around the world is World Bank Open Data. **B**

b: Data from the United Nations and its related agencies is UN Data. **A**

c: The platform that runs data analysis competitions is Kaggle. **C**

EXAM-STYLE QUESTION

Explain why an analyst must check the terms of use and cite the source even when the data is free and open. **[6] Refer to:** your right to use the data, reliability, and letting others verify.

Try

1 Answer the following questions.

1. What is the name of the World Bank's open data portal that publishes statistics on economy, population, education, and so on for countries around the world?
2. What is the term for the rules for how pieces of software interact with each other, which is also widely used in web services for obtaining data?
3. What is the name of the data format widely used when data is returned from an API?
4. What is the term for what should be cited when using data, both to indicate reliability and to allow verification by others?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the flow of obtaining data via an API.

A The client sends a request to the server, and the server returns the data in a format such as JSON.

B The server one-sidedly continues to send data to the client.

C Clients exchange data directly with each other.

D The server only stores data; clients cannot retrieve it.

(2) For each of purposes (a, b), choose the most appropriate open data provider from the choices below (A–C).

a You want to compare GDP across countries.

b You want to find datasets from various fields for machine learning study.

[Choices] **A** UN Data **B** World Bank Open Data **C** Kaggle

(3) From the following options (A–D), choose the one that most appropriately describes the use of open data.

A Open data can be used freely, so the terms of use do not need to be checked.

B Source citation is required only for data published for a fee.

C Even when using open data, it is required to cite the provider, the name of the data, the date of acquisition, the URL, and so on as the source.

D For data obtained via an API, there is no need to follow the terms of use.

PAUSE & THINK

Which provider fits which purpose — and would you use a browser or an API?

Think as an Engineer — Investigate & Decide

For a school project you want to show how Internet access in Egypt has changed over the past ten years compared with two neighboring countries.

You must choose an open data source and a way to obtain it.

☐ **Take the first dataset you find** ☐ **Choose a reliable source and follow its rules**

1 · Inspect two sources. Look up two candidate providers for this exact question (for example World Bank Open Data and UN Data). For each, note whether it actually holds Internet-access figures by country and year, and how current the latest year is.

2 · Analyze fit and rules. Compare the two on a point the lesson table does not list — such as how far back the data goes, or whether its terms of use allow you to republish a chart in a public report. Name one risk of picking the wrong source here.

3 · Decide. Choose one provider and one fetch method (browser download or API) and justify both from what you found. Write the source citation you would include (provider, data name, date obtained, URL). **In pairs,** compare choices and agree on the more defensible source.

Give evidence for your answer.

Stuck? Start from the question, not the source: list exactly what fields you need (country, year, Internet-access rate), then keep only the providers that actually carry all three — and check each one's terms before you commit.

KEY TERMS

terms of use — the rules set for each dataset.
source citation — provider, name, date, URL.

🌍 In a New Context

A weather app on a phone shows today's forecast by fetching it from a public weather service. Describe the request-and-response steps between the app and the server, and name the data format the server most likely returns.

❓ Lesson Question — Answered

The lesson asked where reliable open data comes from, how programs fetch it, and what rules govern its use. Reliable open data comes from trusted providers chosen to fit the purpose — World Bank Open Data for country economic and social statistics, UN Data for United Nations international statistics, and Kaggle for datasets used in machine learning study. It can be downloaded through a browser or fetched by a program through an API: the client sends a request to an endpoint, and the server returns the data in a set format, most often JSON, which the client then uses. Because open data still comes with rules, every dataset carries terms of use that must be followed — including not overloading an API with requests — and every use requires citing the source (provider, data name, date of acquisition, URL), which signals reliability and lets others verify the analysis.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

To use open data in analysis, it is necessary to understand where to obtain it, how to obtain it, and how to use it. Statistics on economy, population, education, and so on for countries around the world can be obtained from sources such as (a), operated by the World Bank. Also, in addition to downloading individually from a browser, by using (b), necessary data can be obtained directly in formats such as (c) in response to a request. When using open data, the (d) determined for each dataset must always be checked, and when data is used, the provider, the name of the data, the date of acquisition, the URL, and so on must be cited as the (e).

1. Fill in the blanks (a)–(e).

⭐ REFLECT & CHALLENGE

Reflect: Think of a chart you saw online with no source. Did you trust it — and what one line would have changed that?
Challenge: You may cite only one source line for a chart built from a World Bank dataset you downloaded today. Write that citation line, and say which single element you would never drop and why.

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes Kaggle.

A A portal operated by the World Bank that publishes statistics for countries around the world.

B A site for international statistics operated by the United Nations.

C A platform that runs data analysis competitions, on which datasets in various fields are made public.

D A system that automatically records browsing histories of websites.

(2) From the following options (A–D), choose the one that is NOT an appropriate description of source citation.

A Source citation is important as an indication of the reliability of the data.

B Source citation is important to allow others to verify the analysis.

C Sources include the data provider, the name of the data, the date of acquisition, the URL, and so on.

D When using open data, it is not necessary to cite the source.

(3) From the following options (A–D), choose the one that is NOT an appropriate point of caution when using an API.

A Use the API in accordance with the terms of use determined by the provider.

B Avoid sending excessive requests in a short time.

C Confirm the scope of allowed use of the obtained data in advance.

D Because of how APIs work, they may be used without checking the terms of use.

⭐ Key takeaway

Remember:
Pick a reliable open data source for your purpose (World Bank Open Data, UN Data, Kaggle), fetch by browser or API (client requests an endpoint; server returns JSON), and always follow the terms of use and cite the source.